

TU257

Information Systems

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Conceptual Design

Initial Requirements

A car rental company needs a new DBMS. The company needs to deal with customers' details, reservations and returns. A customer can book several cars but each reservation is made to only one customer. A fleet of cars is stored at each location and cars have to be picked up and returned to the same location.

Entity identification

The following entities were identified from the requirements:

Car

Reservation

Customer

Relationships

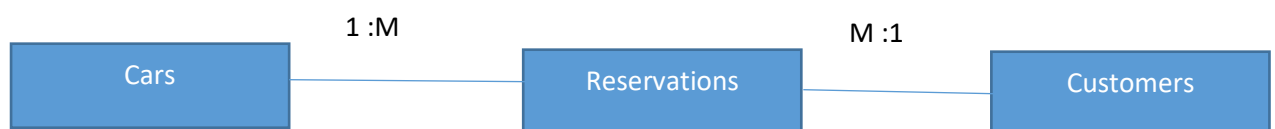
Include relationship matrix or text covering all relationships

Cardinality

The cardinality has been identified as 1 to many between cars and reservations as only one car can appear for each reservation but each car can appear in many reservations.

For customers and reservations it is many to one as one customer can create many reservations but each reservation can only be completed by one customer.

Rough ER diagram



Add main attributes

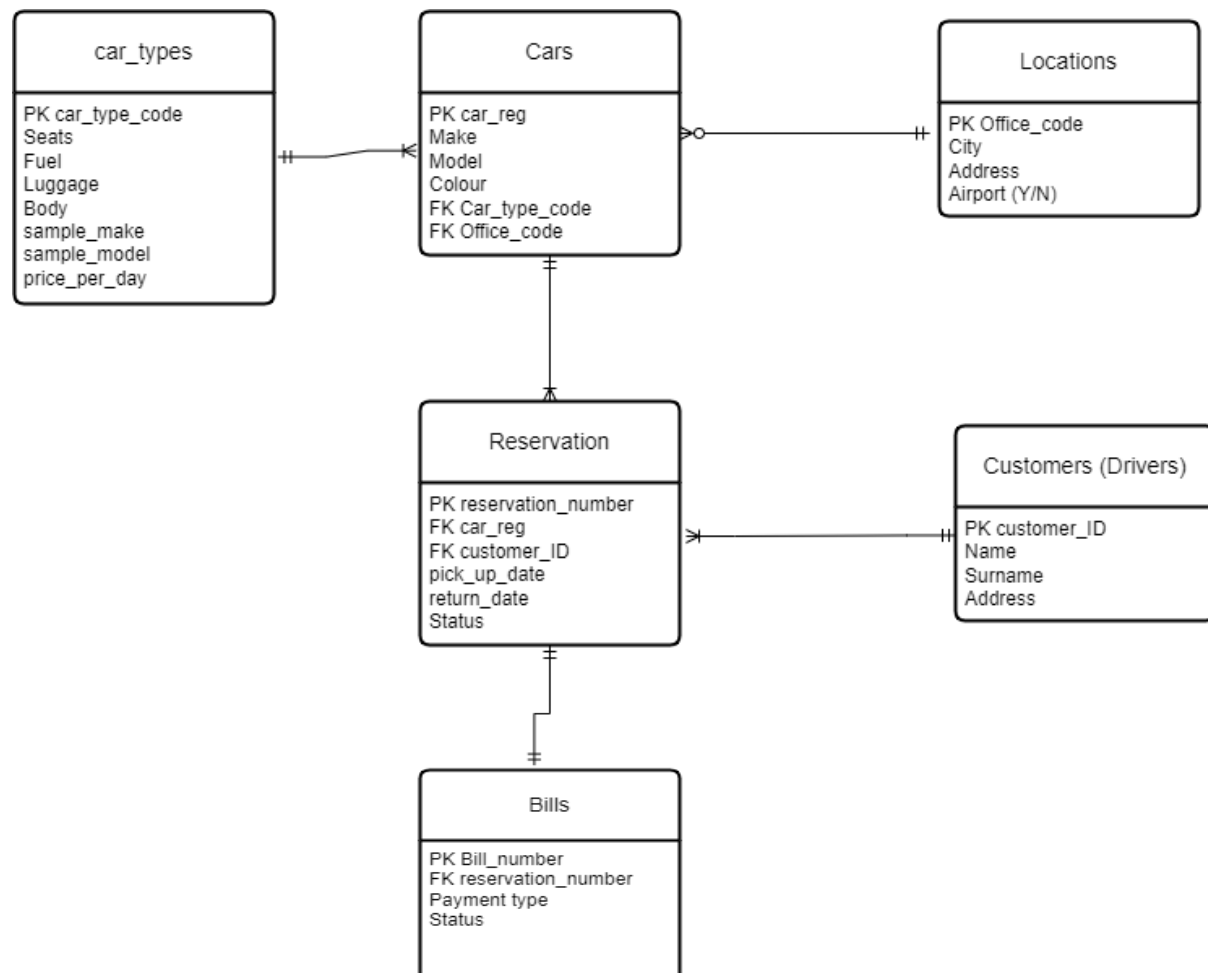
Key based ER Diagram

At this point more entities have been identified, for example car_types as the details and prices per day are determined by the category or type of car. Each car is of one car type and

each car type can have many associated cars. Locations have also been added to store information about each location where cars are stored (picked up and returned to the same location) and bills to identify payments. Each bill is attached to one payment.

Each car is associated with one location but each location can have many cars.

An attribute status is included in the bills entity to record whether the bill has been paid or is pending.

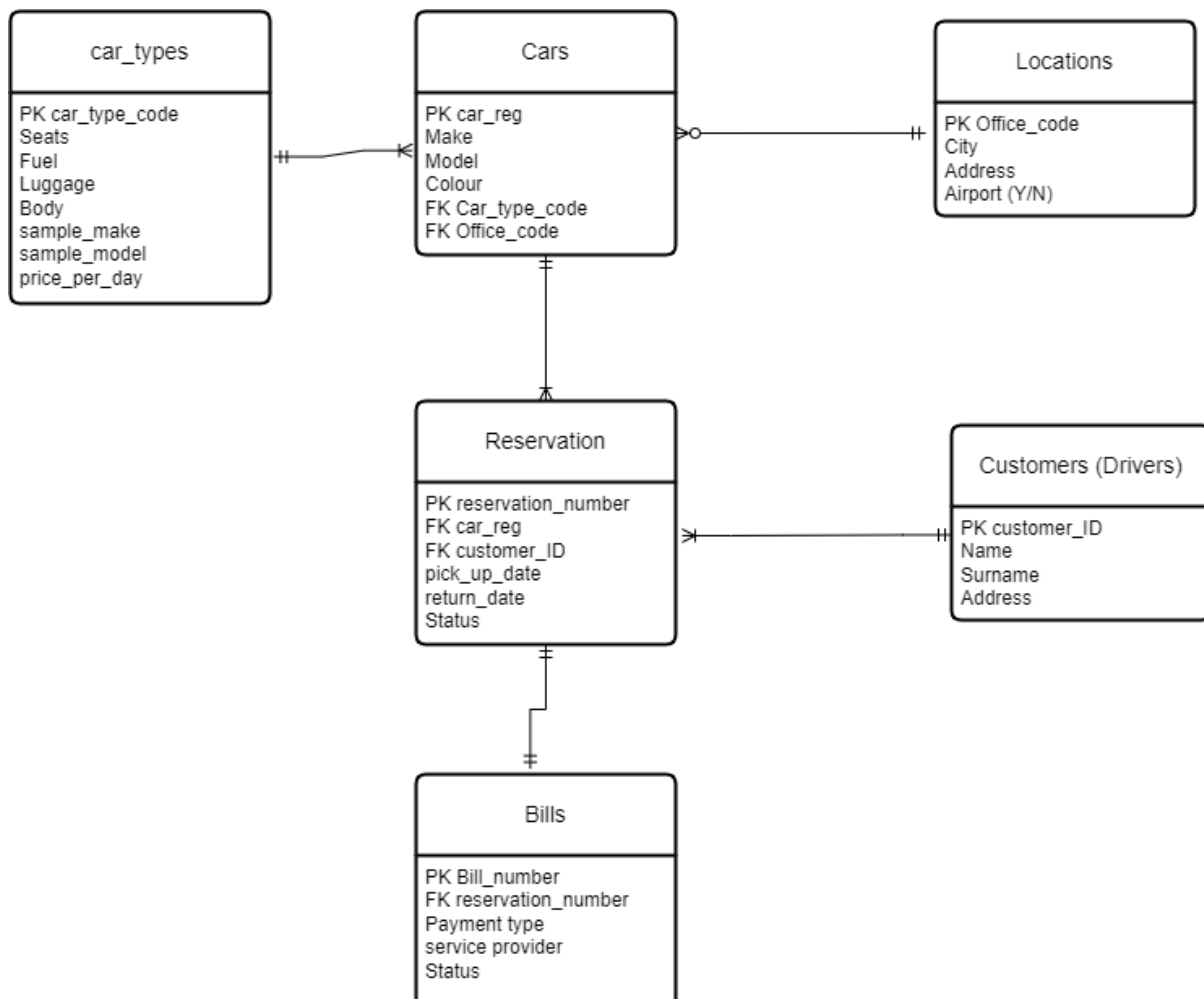


Fully Attributed ER Diagram

A number of extra attributes have been identified as important and necessary, for example driver's license, Driver's DOB, nationality, car_mileage, rental_mileage.

Explain any attribute that may need clarification

Include fully attributed diagram



List of Assumptions

1. Car characteristics are determined by their car type. For each car type there may be a number of cars available.
2. Car types have fixed prices. Sample make and model are fixed per car type.
3. Each location can store a number of cars but cars can only be allocated to one location.
4. Location and customer address are given using the Eircode.
5. Each reservation only contains data about one car and one customer (no extra drivers). Each reservation has a corresponding bill (one) for the full amount.
6. Each bill contains details about one reservation only.
7. Each customer can book different cars on different reservations.
8. Each car can be booked on different reservations.

Make sure all the cardinalities are covered in the requirements or assumption, remember that each relationship has two directions and both need to be explained.

Final considerations

Add some final remarks on the process, limitations or possible improvements of the system designed.

Logical design

Unnormalised Form

CARRENTAL(BookingID, car_type, base_rent, cust_ID, Name, DOB, Age, Cust_address, date_booking, service, service_charge)

Age is a calculated field so it gets removed.

Identification of repeating groups

CARRENTAL(**BookingID**, car_type, base_rent, cust_ID, Name, DOB, Cust_address, date_booking, (service, service_charge)*)

First Normal Form

Repeating groups have been removed and moved to a separate relation with a composite PK that includes the original PK:

CARRENTAL1(**BookingID**, car_type, base_rent, cust_ID, Name, DOB, Cust_address, date_booking)

SERVICES1 (**BookingID, service**, service_charge)

Second Normal Form

Each relation is checked for partial dependencies

CARRENTAL2(**BookingID**, car_type, base_rent, cust_ID, Name, DOB, Cust_address, date_booking)

- No composite key so already in 2NF

SERVICES1 (**BookingID, service**, service_charge)

- Partial dependency found (service_charge depends on service but not on bookingID)

SERVICEBOOKING2 (**BookingID, service**)

SERVICEDETAILS2(**service**, service_charge)

Final 2NF

CARRENTAL2(**BookingID**, car_type, base_rent, cust_ID, Name, DOB, Cust_address, date_booking)

SERVICEBOOKING2 (**BookingID**, service)

SERVICEDetails2(service, service_charge)

Third Normal Form

Each relation is checked for transitive dependencies

CARRENTAL2(**BookingID**, car_type, base_rent, cust_ID, Name, DOB, Cust_address, date_booking)

- Transitive dependencies found for base_rent (car_type) and Name, DOB and Cust_address(cust_ID)

CARRENTAL3(**BookingID**, car_type, cust_ID, date_booking)

CARDETAILS(car_type, base_rent,)

CUSTOMER3(cust_ID, Name, DOB, Cust_address)

- No transitive dependencies found

SERVICEBOOKING3 (**BookingID**, service)

SERVICEDetails3(service, service_charge)

Final 3NF

CARRENTAL3(**BookingID**, car_type, cust_ID, date_booking)

CARDETAILS(car_type, base_rent,)

CUSTOMER3(cust_ID, Name, DOB, Cust_address)

SERVICEBOOKING3 (**BookingID**, service)

SERVICEDetails3(service, service_charge)

List of Assumptions

- Age is calculated using the DOB
- Base_rent depends on the car type
- Each service has a defined price independent from the car type

Make sure all the different dependencies are included here.

Physical Implementation

Creation of tables

Based on the previous design the following tables were created: carsdetails, carrental, etc..

For each table the attributes used are:

Cardetails: cartype, baserent.

For each of these attributes the datatype used was:

cartype varchar(15) as the expected values are text variable length including one word description that fits in 15 characters.

Baserent int as the expected values are numerical integers storing the daily rent rate.

ER diagram

The ER diagram obtained using reverse engineering is:

Add diagram here.

Describe the FKs and PKs.

Queries

Describe the queries (video with the queries running attached to the submission too).

Query one

This query provides information about all reservations made by customer 1.

Query two

....

Optimisation

Based on the given queries and the expected use include optimisation tools and rationale for including it.